Carnegie Mellon University
Computer Science Academy

Design and Development Guide

Before beginning this project, be sure to review the rubric to ensure you include all the required elements.

Description

Describe your program. What do you envision your app will do?

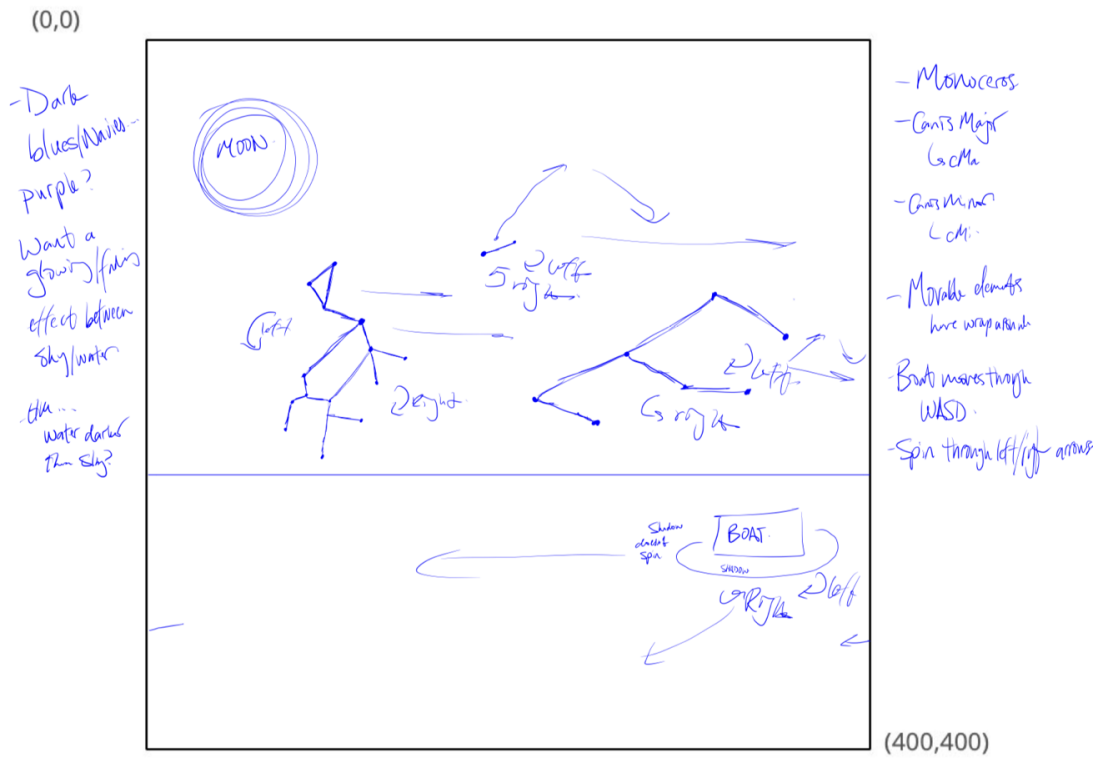
Note: At this step, do not be concerned about how to do it, just focus on what you envision.

A boat on a lake or body of water at night. Full moon hangs in the sky. Either fireworks moving up and exploding or stars growing and decreasing in size and maybe flickering. The boat also moves, either automatically or through key holds.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?



Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
Group	Make it easier to “animate”/move a constellation at once. No one wants to move one star at a time... So through group I can move the constellation while being able to rotate a single star at once. Also helpful for organizing what’s part of what.
onKeyHold/Release	Moves the boat. Hold WASD to move and release to stop. Also hold left/right allows the boat and constellations to rotate counterclockwise or the other way, and releasing it would change them back to the original way.
onStep	When ‘run’ is pressed everything that’s supposed to automatically move starts moving on their own (for this it’s stars). No key holding is needed to make them move.

.dx/dy	Custom properties (yay)! It's for the amount of change in the centerX/Y value. I'll use them so I won't have to scroll up trying to find how much I wanted each shape/group to move I just have to type shape.centerX += shape.dx.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

For this project, originally, I wanted fireworks (and maybe even reflections) in the sky. I remembered there was a similar exercise before, so I tried to find it and try something similar. It did not work. So I thought, 'what else is in the sky at night during new years in a mountain? Aha. Stars", and fireworks became stars. As I was figuring out how to move the constellations after I plotted them out, I suddenly thought of how rotation exists, so I made them spin instead, and liked the look so I kept it but ended it making is so when pressed left/right arrows they'll spin. From my perspective going forward with the stars gave the project more personality... somehow. It's not just a pretty and serene scene. Surprise. The world is spinning. As a result of the spinning stars, I made the boat also spin (but not its shadow).
The more challenging part for the stars was trying to replicate the look of a specific constellation (close enough) as well as deciding what I should do with it. I could either move them in a straight line, not have then move at all, or add in a more rounded path. I tried with the more rounded but it looks more like the constellation's climbing a mountain getting to the top and going down... and I still am not sure how to make it go more smoothly. Making it move slightly up and then down was a challenge that worked, making it go smoother was a challenge that failed.
For trade-offs, I had to scrap the firework idea entirely. They sky would be too much chaos. As for the reflection in water part, I also decided to get rid of that because of the amount of time and ability I had.

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Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
s.add(Star(mouseX,mouseY,2,5,fill=gradient('white','gold'),rotateAngle=-35)) ***the group.add()***	118	.add: The method makes it so that when I click (in this circumstance) the object will become part of that group.	When I click somewhere on the screen, a star gets added to the group "s". This is so that when I do something about this group all the clicked stars will be part of it and it won't interfere with the other constellations. It also breathes more life into the scene – there's more than 3 constellations in the sky.
The group for the Canis major	80-86	Groups the individual child properties together in to a group.	Instead of graphing the stars inside the group, I graphed the stars individually and gave them all names and then added the names to the group. This made it so that I could spin the individual stars separately from the whole group.
s.rotateAngle+=100	195	Rotate angle makes the group spin, and the +=100 spins the group by that value.	It makes the click add-on stars rotate and spin. It's a small thing, but I used it to make a somewhat flickering effect for the stars if it's the right amount (for me it's seven spread out clicks).
if ('left' in keys): canisMajor.rotateAngle -= 0.5	124, 125	If the left arrow key is true then the group will rotate by 0.5.	To make the whole Canis major constellation spin when the left key is pressed. It started off as a random thought and trial that I decided I'd like to keep. It's also the moment I realized I could use decimals.

Future goals

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Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

If I had to change my code, I would add really faint reflections in the water to add more depth in the scene but at the same time take not that much attention away from the sky. At one point I also wanted to add meteorites so it'll be more of "don't get hit and drown" type of interactive scene instead of just spinning the world scene. If I were to somehow add meteorites, I would delete some of the stars and the moon (or maybe have the moon crash on the boat or fill the screen).
During the process, I had to change things as I went because I realized A: I don't have the ability and/or B: there isn't enough time. Whether it be fireworks exploding, smoothly moving the stars and reflections, or meteors landing on the boat, I just couldn't really. So I had to simplify a lot down. But in the simplifying process, I realized I could spin the stars and rolled with that. When planning, I should consider realistic factors in what I want to make and ability to make it...
For resource utilization, I looked at the notes and past exercises, and reached out for help when I needed it. I also got some insight from my friends on how they viewed the project and tweaked some things according to that (though most of those were artistic elements). AI may have been beneficial for the project but not that helpful for my brain... and I don't think I know *how* to use AI either..? Not sure how that works but it doesn't really cooperate.
As for time management, I could have spent less time changing the tiny details such as how the shadow should look and focused more on the technical aspects in the time that we had, since so much of my time during these projects goes to designing how it should look rather than how it should act.